Stoichiometry

30 questions: 15 multiple choice, 15 structured.

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1 Core Formulae [1]

Limestone is mainly calcium carbonate. Which formula represents calcium carbonate?

A Ca_2CO_3

B $CaCO_2$

C $CaCO_3$

D $Ca(CO_3)_2$

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2 Core Formulae [1]

What does the molecular formula of a compound show?

A The number and type of each atom in one molecule of the compound

B The simplest whole-number ratio of atoms in the compound

C The charges on the ions in one formula unit of the compound

D The percentage by mass of each element in the compound

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3 Core Formulae [1]

Magnesium ribbon burns in air to form magnesium oxide. Which symbol equation, including state symbols, correctly represents this reaction?

A $Mg(s) + O_2(g) \rightarrow MgO_2(s)$

B $Mg(s) + O(g) \rightarrow MgO(s)$

C $2Mg(s) + O_2(g) \rightarrow MgO(s)$

D $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

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4 Extended Formulae [1]

A compound has the molecular formula C_4H_{10}. What is its empirical formula?

A C_4H_{10}

B C_2H_5

C CH_2

D C_2H_{10}

5 Extended Formulae [1]

Aluminium ions, Al^{3+}, and sulfate ions, SO_4^{2-}, combine to form an ionic compound. What is the formula of this compound?

A $Al_2(SO_4)_3$

B $AlSO_4$

C Al_2SO_4

D $Al_3(SO_4)_2$

6 Extended Formulae [1]

Aqueous silver nitrate is added to aqueous sodium chloride, forming a white precipitate. Which ionic equation, with state symbols, correctly represents this reaction?

A $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

B $Ag^+(aq) + NO_3^-(aq) \rightarrow AgNO_3(s)$

C $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

D $2Ag^+(aq) + Cl^-(aq) \rightarrow Ag_2Cl(s)$

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7 Core Formulae [4]

Sodium reacts with water to produce sodium hydroxide and hydrogen gas.

(a) Give the formula of each of the following:

(i) sodium hydroxide [1]

(ii) hydrogen gas [1]

(b) Write a balanced symbol equation for this reaction, including state symbols. [2]

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8 Core Formulae [4]

The table shows the number of each type of atom present in one molecule of three compounds, P, Q and R.

Compound	Carbon atoms	Hydrogen atoms	Oxygen atoms
P	1	4	0
Q	2	6	1
R	1	2	1

(a) Deduce the molecular formula of compound P. [1]

(b) Deduce the molecular formula of compound Q. [1]

(c) Deduce the molecular formula of compound R. [1]

(d) State the meaning of the term *molecular formula*. [1]

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9 **Core** Formulae [4]

Calcium carbonate is a solid. When it is heated strongly in an open container, it decomposes into calcium oxide, which is also a solid, and carbon dioxide gas.

(a) Write a word equation for this reaction. [1]

(b) Write a balanced symbol equation for this reaction, including state symbols. [2]

(c) Explain, using the equation, why the total mass of solid remaining decreases when calcium carbonate is heated in an open container. [1]

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10 **Extended** Formulae [4]

A hydrocarbon molecule contains 6 carbon atoms and 12 hydrogen atoms.

(a) State the meaning of the term *empirical formula*. [1]

(b) Give the molecular formula of this hydrocarbon. [1]

(c) Deduce the empirical formula of this hydrocarbon. [1]

(d) A different hydrocarbon has the empirical formula CH_2 and its molecule contains 4 carbon atoms. Deduce its molecular formula. [1]

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11 **Extended** Formulae [5]

The table gives the formula and charge of four ions.

Ion	Formula	Charge
Barium	Ba	2+
Sulfate	SO_4	2−
Chloride	Cl	1−
Sodium	Na	1+

(a) Deduce the formula of barium sulfate. [1]

(b) Deduce the formula of sodium chloride. [1]

(c) Aqueous barium chloride is mixed with aqueous sodium sulfate. A white precipitate of barium sulfate forms, leaving sodium chloride in solution. Write a balanced symbol equation for this reaction, including state symbols. [3]

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12 **Extended** Formulae [5]

Dilute hydrochloric acid reacts with aqueous sodium hydroxide in a neutralisation reaction, producing water and aqueous sodium chloride.

(a) Write a full balanced symbol equation for this reaction, including state symbols. [2]

(b) Identify the two spectator ions in this reaction. [1]

(c) Write the ionic equation for this neutralisation reaction, including state symbols. [2]

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13 **Core** Relative masses of atoms and molecules [1]

Which statement correctly defines relative atomic mass, Ar?

A The mass of one atom of an element measured in grams

B The average mass of the isotopes of an element compared with 1/12 of the mass of an atom of carbon-12

C The number of protons and neutrons in the nucleus of an atom

D The mass of one mole of atoms of an element

14 Core Relative masses of atoms and molecules [1]

Using Ar: C = 12, O = 16, Ca = 40, what is the relative formula mass, Mr, of calcium carbonate, $CaCO_3$?

A 68

B 84

C 116

D 100

15 Core Relative masses of atoms and molecules [5]

Ar: H = 1, C = 12, N = 14, O = 16.

(a) Calculate the relative molecular mass, Mr, of carbon dioxide, CO_2. [1]

(b) Calculate the relative molecular mass, Mr, of glucose, $C_6H_{12}O_6$. [2]

(c) Calculate the relative formula mass, Mr, of ammonium nitrate, NH_4NO_3. [2]

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16 Core Relative masses of atoms and molecules [4]

In the reaction $2Mg + O_2 \rightarrow 2MgO$, 24 g of magnesium reacts completely with oxygen to form 40 g of magnesium oxide (relative formula mass of MgO = 40).

(a) Calculate the mass of oxygen that reacts with 24 g of magnesium. [1]

(b) Calculate the mass of magnesium oxide formed when 9.6 g of magnesium is completely oxidised, using simple proportion. [2]

(c) Calculate the mass of oxygen needed to react completely with 9.6 g of magnesium. [1]

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17 Core Relative masses of atoms and molecules [4]

Copper carbonate decomposes on heating: $CuCO_3 \rightarrow CuO + CO_2$

The relative formula masses are: $CuCO_3 = 124$, $CuO = 80$, $CO_2 = 44$.

(a) Show that these three relative formula masses are consistent with conservation of mass in this equation. [1]

(b) 3.1 g of copper carbonate is heated until it decomposes completely. Calculate the mass of copper oxide formed. [2]

(c) Calculate the mass of carbon dioxide gas released in part (b). [1]

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18 Core The mole and the Avogadro constant [1]

Which pair of units can both be used to express the concentration of a solution?

A g/dm^3 and mol/dm^3

B g/cm^3 and mol/g

C mol/dm^3 and dm^3/mol

D g/mol and mol/dm^3

19 Extended The mole and the Avogadro constant [1]

What is the value of the Avogadro constant, and what does it represent?

A 6.02 × 10^23; the number of moles in one gram of a substance

B 6.02 × 10^-23; the number of particles in one mole of a substance

C 6.02 × 10^23; the number of particles in one mole of a substance

D 6.02 × 10^23; the mass, in grams, of one mole of a substance

20 Extended The mole and the Avogadro constant [1]

Ar: Na = 23. What amount, in moles, is present in 11.5 g of sodium?

A 0.5 mol

B 0.05 mol

C 2.0 mol

D 5.0 mol

21 Extended The mole and the Avogadro constant [1]

The molar gas volume is 24 dm^3 at room temperature and pressure (r.t.p.). What is the volume occupied by 0.25 mol of a gas at r.t.p.?

A 24 dm^3

B 96 dm^3

C 0.25 dm^3

D 6 dm^3

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22 **Extended** The mole and the Avogadro constant [1]

0.25 mol of solute is dissolved in 250 cm^3 of solution. What is the concentration of this solution, in mol/dm^3?

A 0.1 mol/dm^3

B 1.0 mol/dm^3

C 4.0 mol/dm^3

D 1000 mol/dm^3

23 **Extended** The mole and the Avogadro constant [1]

A compound contains 82.4% nitrogen and 17.6% hydrogen by mass. Ar: N = 14, H = 1. What is the empirical formula of this compound?

A NH

B NH_2

C NH_3

D N_3H

24 **Extended** The mole and the Avogadro constant [1]

A student expected to obtain 12.0 g of a salt from a reaction but actually obtained 9.6 g. What is the percentage yield?

A 20%

B 80%

C 96%

D 125%

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25 **Extended** The mole and the Avogadro constant [5]

Ar: Na = 23, O = 16, H = 1.

A solution of sodium hydroxide has a concentration of 8.0 g/dm^3.

(a) Calculate the relative formula mass, Mr, of sodium hydroxide, NaOH. [1]

(b) Calculate the concentration of this solution in mol/dm^3. [2]

(c) Calculate the amount, in moles, of sodium hydroxide present in 250 cm^3 of this solution. [2]

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26 **Extended** The mole and the Avogadro constant [5]

Ar: Mg = 24.

(a) Calculate the amount, in moles, of magnesium atoms in 3.0 g of magnesium. [2]

(b) Calculate the number of magnesium atoms in this 3.0 g sample. [2]

(c) Calculate the mass of 0.60 mol of magnesium atoms. [1]

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27 **Extended** The mole and the Avogadro constant [7]

Ar: Mg = 24, H = 1, Cl = 35.5.

Magnesium reacts with hydrochloric acid: $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

2.4 g of magnesium is added to 120 cm^3 of hydrochloric acid of concentration 2.0 mol/dm^3.

(a) Calculate the amount, in moles, of magnesium used. [1]

(b) Calculate the amount, in moles, of hydrochloric acid used. [2]

(c) Determine which reactant is in excess. Show your working. [2]

(d) Calculate the maximum mass of magnesium chloride, $MgCl_2$, that could form. [2]

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28 **Extended** The mole and the Avogadro constant [4]

Ar: Na = 23, C = 12, O = 16.

Excess dilute sulfuric acid reacts with 5.3 g of sodium carbonate:

$Na_2CO_3(s) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + H_2O(l) + CO_2(g)$

(a) Calculate the amount, in moles, of sodium carbonate reacted. [1]

(b) Calculate the amount, in moles, of carbon dioxide gas produced. [1]

(c) Calculate the volume of carbon dioxide gas produced, measured at r.t.p. [2]

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29 **Extended** The mole and the Avogadro constant [6]

In a titration, 25.0 cm^3 of a sodium hydroxide solution of unknown concentration was exactly neutralised by 22.5 cm^3 of hydrochloric acid of concentration 0.100 mol/dm^3.

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

(a) Calculate the amount, in moles, of hydrochloric acid used. [1]

(b) Calculate the amount, in moles, of sodium hydroxide that reacted. [1]

(c) Calculate the concentration of the sodium hydroxide solution, in mol/dm^3. [2]

(d) Ar: Na = 23, O = 16, H = 1. Calculate the concentration of the sodium hydroxide solution in g/dm^3. [2]

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30 **Extended** The mole and the Avogadro constant [5]

A hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass. Its relative molecular mass, Mr, is 42.

Ar: C = 12, H = 1.

(a) Calculate the empirical formula of this hydrocarbon. Show your working. [3]

(b) Calculate the molecular formula of this hydrocarbon. [2]

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